IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 24 May 2025, accepted 18 June 2025, date of publication 27 June 2025, date of current version 9 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3583929

RESEARCH ARTICLE

Planning and Scheduling System for Electric Vehicle Charging

PIOTR PAŁKA[1], TOMASZ ŚLIWIŃSKI[1], PRZEMYSŁAW KASZYŃSKI[2], MARTA KUTA[3], BOGDAN RUSZCZAK[4], (Member, IEEE), MARCIN MALEC[2], PIOTR SAŁUGA[5], AND JACEK KAMIŃSKI[2]

[1]Institute of Control and Computation Engineering, Warsaw University of Technology, 00-665 Warsaw, Poland
[2]Mineral and Energy Economy Research Institute, Polish Academy of Sciences, 31-261 Kraków, Poland
[3]Faculty of Energy and Fuels, AGH University of Krakow, 30-059 Kraków, Poland
[4]Faculty of Computer Science, Opole University of Technology, 45-758 Opole, Poland
[5]Faculty of Applied Sciences, WSB University, 41-300 Dąbrowa Górnicza, Poland

Corresponding author: Bogdan Ruszczak (b.ruszczak@po.edu.pl)

This work was partially supported by The National Centre for Research and Development of Poland under project POIR.01.01.01-00-0855/21. It was also carried out as part of the statutory research activity of the Mineral and Energy Economy Research Institute of the Polish Academy of Sciences. This research was partially supported by the Ministry of Science and Higher Education, Poland, AGH grant no. 16.16.210.476.

ABSTRACT The paper presents a novel approach to instant on-the-road electric vehicle charging scheduling planning. It benefits from a mixed-integer mathematical optimization model employed to efficiently select charging stations along the route. It also utilizes machine learning models to approximate the traction battery state of charge using data on driving conditions (energy, speed, travel time, etc.) for individual route sections, considering the driver's habits and preferences. The research details the functional requirements of this procedure application and its architecture. A step-by-step description of the optimization model is provided, in particular, the constraints responsible for the non-linear model of charging time, the varying time zone-dependent trip parameters (including rush hours), as well as the used optimization criteria, and the scalarization methods. Unlike other existing methods, the combination of the aforementioned mixed-integer mathematical optimization model and machine learning predictors allows for consideration of several factors and adaptation to actual data. Experiments are carried out on the operation of the application, including very long routes (with the actual Warsaw-Zakopane, 412 km long test drives). The results show enhanced, satisfactory execution times, from 0.088 [s] (for a simple model without time zones and with a linear charging model) to 0.365 [s] (for a model without time zones and with a non-linear charging model).

INDEX TERMS Electric vehicles, electromobility, EV, machine learning, mixed-integer programming, scheduling.

I. INTRODUCTION

The wide adaptation of electric vehicles (EV) we observed in recent years raises a whole area of novel studies. There is a growing need for many optimizations. The charging network has to develop and become better integrated and accessible. The network needs to boost participation in electric power markets to allow the energy market and ancillary service market to enhance renewable energy inclusion [1]. The location of charging stations is another researched area of optimization, with special analysis for future station deployment or reallocation of existing ones [2].

The easily accessible, located close to the end user, smart workplace parking garages should be popularized [3]. Also, to better understand the battery recharging trends and to improve the distribution of the charging stations, we should study different clients' strategies [1] or work on enhancing various battery technologies [4], that could potentially allow for longer discharge times or less battery cell degradation, reduced battery cell degradation, and a slower battery aging

The associate editor coordinating the review of this manuscript and approving it for publication was Amin Mahmoudi.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

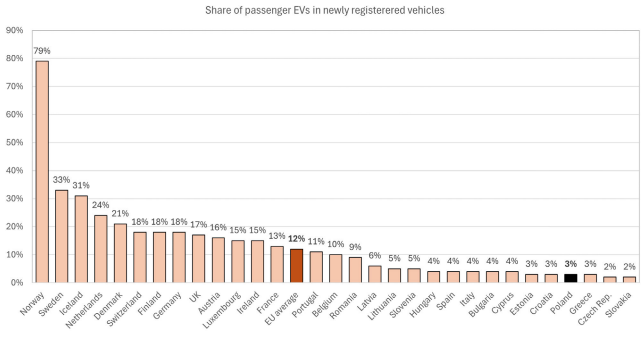

FIGURE 1. Share of passengers for newly registered EVs in 2022 [20].

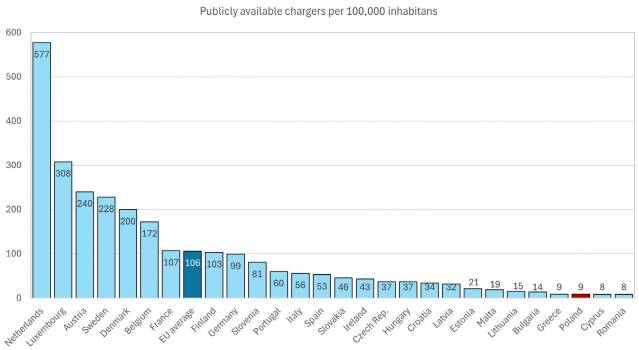

FIGURE 2. Publicly available chargers per 100,000 inhabitants, 2022 [19].

process [5]. As stated in [6], the optimization of the charging process could also be beneficial and could be simulated for a number of scenarios using the proposed framework. Also, as it is reported by some researchers [5], [7], the vehicle's state of charge is also highly related to the user's recharging behavior.

One of the tasks currently tackled is recharging scheduling optimization, which is widely investigated [8], [9], [10], [11], [12], [13], [14], [15]. This task is somewhat straightforward - it aims to propose when or where to recharge the vehicle. But when the options the client could consider are taken into account, it becomes complex instantly. One of the currently proposed frameworks [13] points out the potential of personalized individual driver requirements that could be included in the planning. They utilize both the traffic information and the individual vehicle data to obtain an equilibrium solution iteratively. The other one proposes coordinating three types of charging that could lead to the optimal charging dispatch [8], also considering charging rates [11]. Some other researchers advise introducing the data of various driving and terrain scenarios (tested on Colombian routes on an airport shuttle service scenario) and related driving phenomena (the battery degradation) [14]. Improving the energy efficiency of movement patterns in urban road traffic using navigation satellite data [16] as well as optimizing traffic using spatio-temporal floating vehicle data for logistics operations [17] are also widely considered, as well as employing deep learning models for similar tasks, that is reported as beneficial [18]. The upcoming rise in demand driven by EVs will enforce the need for effective demand management [9], and could possibly be an important factor for scheduling applications.

For the later presented approach, we also follow the digital twin (DT) methodology for the optimization task, as also applied in [13], but we set different objectives (fewer charging steps, minimized detours for accessing charging infrastructure, and enhanced economic efficiency). Some researchers reported Ant Colony Optimisation, among other optimization techniques, as an effective tool for planning fixed route vehicle charging problems [15], with decisions on the recharging schedules of the routes included in the analyzed task.

Despite the increase in the number of EVs and charging stations in Poland, the market is still in the early stage of development. Both the share of newly registered electric cars and the number of publicly available charging stations per 100,000 inhabitants are significantly below the average for European countries [19] (see Figure 1 and Figure 2). The possible lack of access to charging stations may concern not only EV owners from Poland but also most European countries.

According to the survey (Barometer of New Mobility 2020/21, Polish Alternative Fuels Association [21]), Poles face uncertainties that are associated with traveling using an electric vehicle. One of the main barriers (apart from the high price of EVs) is the lack of guarantee of the availability of charging stations (due to, e.g., a small number of charging stations outside large cities or the unavailability/failure of the charger located near the route). Another significant obstacle is the discomfort associated with long charging times (depending on the charging station type).

The consequences of the development of electromobility include resource management, appropriate deployment of supporting infrastructure, and travel planning [21]. However, the lack of sufficient infrastructure and appropriate IT tools to support drivers in planning electric car journeys, taking into account the actual energy consumption of the vehicle (depending on the driver's driving style, weather, road conditions, or route profile) and the selection of charging stations in the vicinity of the planned route and meeting specific criteria, may constitute one of the main barriers to the development of the electromobility market in countries such as Poland. Additional problems related to the everyday operation of EV include both:

1) the uncertainty related to the estimation of the vehicle charging time (this relationship is non-linear and depends on several factors, such as the type of charger and the connected battery/vehicle, battery temperature and ambient temperature), and
2) the uncertainty related to the availability of the selected charging station (the charger may be occupied or out of order) and the waiting time for charging.

One of the challenges that currently affects the use of electric cars is the long charging time and relatively low

range (compared to vehicles with combustion engines). While short-distance travel, such as daily commuting or shopping trips near the place of residence, is convenient for the owner, longer journeys can be more challenging. Proper planning of stops along the route during which the car will be charged is an interesting research topic. For this reason, the integrated model presented in this paper may offer a solution to the challenges posed by the lack of appropriate tools for supporting drivers in the journey planning process. The model combines machine learning (ML) methods with mathematical programming. ML methods are used to accurately predict the range of an electric vehicle for a selected route, considering travel times and rejoining the route after stopping at identified charging stations located on or near the route. Mathematical programming is incorporated to optimize the selection of charging stations and the EVs' charging schedule, accounting for different time zones during which the journey may occur.

For this reason, the integrated model presented in this paper may offer a solution to the lack of appropriate tools supporting drivers in journey planning. The model combines ML methods to accurately predict the range of an electric vehicle for a given route—taking into account travel times and detours to charging stations located along or near the route—with mathematical programming techniques to optimize the selection of charging stations and the EVs charging schedule, including various time windows in which the journey may occur.

Current market trends and user expectations create the need for personalized mobile applications to facilitate and support the user in various daily activities. Calendar apps help efficiently organize daily schedules, while navigation apps guide users to their destinations without requiring precise knowledge of their location. In this context, it is worth considering the design and implementation of a comprehensive application for planning electric vehicle charging schedules along a given route, taking into account both driving conditions and user preferences. This article outlines the key scientific challenges associated with developing such an application. However, the technical aspects of the app's design and implementation are beyond the scope of this paper. Calendar apps make it easier for us to plan our day, and navigation apps allow us to get to a destination without having to know where we are. Consideration should be given to the design and implementation of a complete application to prepare a charging schedule along the car's route, taking into account the driving conditions and user preferences. The article will outline the key scientific challenges used to create the application in question. The technical process of designing the app itself will not be described.

The research hypothesis examined in this article in this article is whether it is possible to create a functional personal mobile application that enables the optimal selection of charging stations, charging level on a fixed route with technical limitations, and a user-defined set of preferences. The components needed to produce the application will also be defined and described in detail.

The following sections of the manuscript include the general description of the proposed method (section II) and the literature review of the existing solutions (section III). The core of the proposed approach consists of two main components: machine learning model, used to accurately estimate changes in the electric vehicle's battery state of charge (described in Section IV), and the optimization model (Section V), which provides the end-to-end solution for the problem of charging an electric car and navigating through the planned road. These descriptions are followed by the elaboration of computational experiments and field tests of the proposed methods (Section VI) and final remarks (Section VII).

II. PROPOSED APPROACH

In order to create a concept for a tool to support the charging of electric cars on the route, the requirements for the application must be formulated. The application must have information on charging stations, including their location, charging rates, charging standards, and additional information on the charging station services (e.g., ATM, restaurant, etc.). The application must have information on the habits that define the driver: typical speed, acceleration, and energy consumption. The application must approximate speed, energy consumption, travel time, etc., on each section of the route; return the optimum charging schedule under given user preferences; provide several optimization criteria, such as: (i) minimization of the total cost of charging; (ii) minimization of the total route travel time; (iii) minimization of the number of stops; (iv) minimization of the risk of prolonging the journey; (v) maximization of driver satisfaction with the currently required facilities (ATM, restaurant, hotel, shop) at charging stations indicated by the system. It should be emphasized that the list of user preferences and station amenities is neither exhaustive nor the result of detailed research. Further work in this area will be necessary during the development and implementation of the final application for market deployment. One of the questions that this article seeks to answer is whether it is feasible to develop mathematical models or tools that consider user preferences when selecting electric vehicle charging stations. Also, the application must allow for parameterization of the optimization model and the share of optimization criteria. The application will not determine the exact route. Instead, the route will be obtained from a dedicated tool.

Figure 3 shows the diagram of the data sources (transparent blocks), computing modules (light blue blocks), and results (dark blue block). In addition, arrows mark the data flow and the order in which calculations are run in the system (see Figure 3). It can be seen that the weather data, road profile, technical parameters of the battery, initial state of charge (SOC), and drivers' habits go to the ML module. The ML module returns the energy consumption of the car on the specific road fragments. This data, together with the

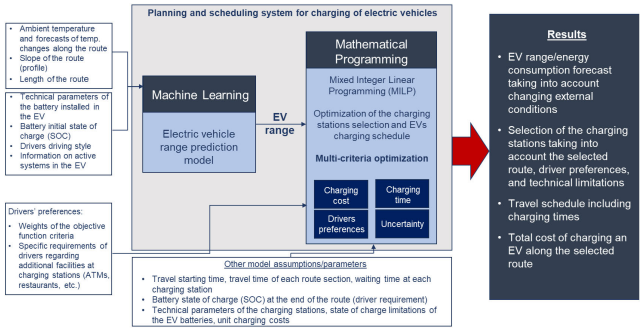

FIGURE 3. Systems' data flow, based on the functional requirements.

driver's preferences and the parameters defining the driving conditions, is passed to the Mathematical Programming module. This model returns the optimal charging schedule in terms of the set criteria at the selected charging stations.

The system we are considering can be divided into three main components:

- Input data preparation: the task of this component is to prepare data into the appropriate formats from external tools and services, such as: short-term weather forecast (weather services), lengths, grades, gradients of route sections, technical parameters of the car, and drivers' habits.
- Data estimation using ML: the model provides predictions of ranges of the vehicle under specific parameters (mentioned above).
- Optimization model: optimizes the schedule of charging on specified routes.

The components are described in detail in the following chapters.

III. LITERATURE AND EXISTING SOLUTIONS OVERVIEW

The state-of-the-art literature, especially in recent years, shows that a great deal of interest has been devoted to the problem of scheduling electric vehicle charging. The works of Chen et al. [22] and Sun et al. [23] described methods of determining the charging schedule for electric cars from the point of view of balancing demand in the energy network, taking into account the preferences of car users [22] and taking into account the road network [23]. Similar issues were presented by Yin et al. [24] and Yin and Ji [25], where the charging schedule for EV was optimized (minimizing network losses) and considering the stable operation of the distribution network in a given area. In both works, predictive models (including the fusion model, LSTM, and XGBoost) were used to determine accurate charging load forecasting.

Studies on optimizing the deployment and location of charging stations are another area of research explored in the literature. In the work of Song et al. [26], a combination of ML methods (a gradient boosting-based learning model is used) and optimization (a demand-supply coupled CSLP model) was proposed to predict charging demand and optimize the load of charging stations (both existing and new/planned).

The location of electric vehicle charging stations, along with the indication of the number and electrical capacity of chargers, was also the subject of the work by Celik et al. [27]. The authors used a genetic algorithm to solve the problem of station location and a simulation tool to model the traffic intensity at the station and optimize the types and number of chargers. One of the decision criteria was to minimize waiting times for charging and the travel distance to the station.

The problem of developing electric vehicle charging infrastructure was also addressed from the perspective of private investors to maximize their revenues from providing electric vehicle charging services. Li et al. [28] presented a three-level location model using a dynamic pricing mechanism that includes user decisions, charging station prices and fees, and decisions related to the location of new stations. The authors also used a reinforcement learning algorithm to develop/find the most advantageous (generating the highest revenues for a given private company) pricing strategy for new charging stations.

Location of charging stations and EV routing can also be treated as interlinked problems. For example, Fathollahi et al. [29] propose a hierarchical decision-making approach, where strategic-level decisions address the location problem, while tactical-level decisions optimize EV routes. The simultaneous solution of these two levels is formulated as a location-routing problem (LRP), incorporating a multi-depot, multi-product, capacitated vehicle routing problem (VRP) and a power distribution-constrained location problem.

Most recent research in the field of charging EV while traveling focuses on the development of methods and algorithms for determining the fastest route between two locations, taking into account the necessary stops for charging the vehicle, taking into account the large range of uncertainty related to the availability of the selected charging station as well as the waiting time for the vehicle charging process to start. This type of problem was described in the work of Dastpak et al. [30], in which the authors proposed a re-optimization algorithm to solve the Markov Decision Process decision problem called the Electric Vehicle Shortest Path Problem (EVSPP). In their research, in order to eliminate the uncertainty related to waiting time, the authors added a binary parameter indicating the availability (or lack of availability) of a given charging point on a real-time basis. The cited work also uses a simulation approach to estimate the waiting time for charging at subsequent necessary stops during the journey based on the arrival time to a given station. This time is estimated using selected heuristics. Dastpak et al. [30] in their research also included the calculation of vehicle charging time using piecewise linear charging functions, depending on the type of charging station technology. Planning a trip for an electric car, including stops at charging stations, using a heuristic algorithm to solve the problem defined as EVSPP was also the subject of the work by Cubillos et al. [31]. In their method, the authors also

included the possibility for users to specify time windows in which selected activities should take place during the trip and other preferences related to preferred points of interest located close to the charging stations.

Kullman et al. [32] proposed a solution to the vehicle routing problem, including the decision to stop the vehicle for charging (defined as a fixed route vehicle charging problem). For this purpose, the authors used a developed tool/algorithm called *frvcpy*, an open-source Python package. The inspiration to develop the mathematical model presented in the paper, used to optimize the scheduling of charging EV over a fixed travel route, also came from work on the scheduling of refueling of a vehicle fleet, taking into account travel distances and fuel consumption. The works of Nourbakhsh and Ouyang [33] and Raviv and Kaspi [34] describe mathematical models that optimize the refueling of a fleet of locomotives moving in the railway network.

The novel approach proposed in our work to solve the decision-making problem related to the optimal selection of electric vehicle charging stations on a fixed travel route includes a combination of two methods, i.e:

1) ML (forecasting the vehicle's energy consumption on individual sections of the route depending on a number of parameters: vehicle type, driver characteristics, weather conditions, route slope, etc.), and
2) mixed-integer linear mathematical programming with multi-criteria objective function (optimization of charging station selection).

The decision to employ Mixed-Integer Linear Programming (MILP) for the planning and scheduling of electric vehicle (EV) charging in our study was driven by several key factors, including the need for optimality, deterministic guarantees, and compatibility with the problem's structure. The EV charging scheduling problem involves discrete decisions and continuous variables, which naturally align with the MILP framework. MILP enables precise formulation of the problem with constraints such as charger availability, battery capacity, and user preferences, while optimizing a multicriteria objective function. Unlike metaheuristic methods, which are probabilistic and may converge to local optima, MILP guarantees global optimality for linear problems when a solution is found, provides deterministic results, and offers flexibility in modeling constraints. Although metaheuristics are generally faster for large-scale problems due to their guided search through the solution space, the MILP runtimes for the use cases considered here are sufficiently low, with a maximum of 30 seconds in the most complex case (Table 2).

In addition to the technical limitations of the vehicle batteries and the chargers installed at the charging stations, the model takes into account the time slots in which the journey takes place (they influence the road and weather conditions and thus the predicted energy consumption), a complex function for estimating the charging time of the vehicle battery (piecewise linearization of a non-linear function) depending on the charging curve of a given vehicle at a given charger, the ability of the user to specify the start time of the journey and additional preferences related to the availability of selected facilities (such as shops, ATMs, hotels, restaurants) close to the charging stations. The use of a multi-criteria optimization, where many criteria that do not necessarily conflict with each other, allowed all criteria to be considered in the optimization model, including criteria to be minimized: travel time, vehicle charging and energy consumption costs, charging waiting times uncertainty, and criterion to be maximized: user preferences regarding availability of facilities near selected charging stations.

The combination of ML methods and MILP is used in many areas. MILP as a tool for interpretability and optimization of ML models, e.g., for optimizing ML hyper-parameters, especially when they have to satisfy discrete constraints [35]. Accelerating and supporting the solution of difficult MILP optimization problems using ML models, e.g., for predicting the values of integer variables. In [36] a Graph Neural Network was used to predict the values of binary variables in the MILP problem. In another work [37], a cut prediction in Branch-and-Cut algorithms using supervised learning and deep reinforcement learning was proposed. There is a growing body of work in the literature investigating how machine learning (ML) algorithms can generate, generalize, or feed data into Mixed-Integer Linear Programming (MILP) optimization models to improve their performance, scalability, or accuracy. Our approach can be described as MILP parameter prediction from partial data using ML. In [38], a description of an OPF-AC model is provided, for which missing parameters, e.g., energy demand, are predicted using neural networks. The authors of [39] propose an Energy-Based Model (EBM) that learns to map incomplete data to the full parameters of an optimization problem. Existing research on ML-based parameter prediction for MILP faces several key bottlenecks: data efficiency and generalization, computational overhead, lack of data explainability, time-varying parameters and noise measurements, and integration with solvers.

Since the currently available systems are rather a combination of a map of charging stations with the information on available chargers of some specific operators, the proposed optimization method would allow for including many more dimensions in the decision scheme. We have proposed an efficient method that incorporates the existing information on the route, charging station viability, costs, among others, with the predictive ML models that fill in the missing state of charge of the EV along the planned route.

IV. ESTIMATION OF CAR ENERGY CHANGES USING MACHINE LEARNING MODELING

The main optimization application requires the precise estimation of the amount of energy that would be consumed or recovered for the planned road. This has to be computed for each road segment and later analyzed by the application. This means the changes in the motor battery's state of charge need to be precisely predicted. The developed regression

model, using the input data, should reflect the energy consumption and recuperation (linked to the commonly equipped regenerative braking subsystems). This information must be up to date – it must relate to the current time when the model is executed and synchronized with the weather situation, the road characteristics (i.e., different for uphill or downhill driving), and the traffic in the area. To deliver a model, that could incorporate all these factors without the need for endless manual modeling of all the hidden data interconnections, we decided to evaluate several ML methods. To achieve this, as suggested in [40], [41], and [42], we applied several different ML model architectures from the group of supervised regression ML techniques, namely: linear models with L1 and L2 regularization [43], regressors based on support vector machines (SVR), ensemble learning methods [44], [45]: random forest (RF), extra trees (ET) and extreme gradient boosting (XGB) [46], and fully connected neural networks (FCNN).

We employed rather straightforward algorithm configurations, with a maximum number of estimators set at 500 for the ensemble methods (ET, RF, XGB). The regularization terms were exclusively tuned for linear models (for L1 and L2 regressors as well as for the SVR with linear kernel). The FCNN is configured with the following settings: Relu activation functions, Adaptive Moment Estimation optimizer and, besides the input and output layers, 5 additional hidden layers (after some tuning we found the network having the following configuration of layers: [300, 250, 200, 150, 100, 50] of best performance).

The employed dataset was built on top of the hundreds of test drives. This initial set consisted of around 26,000 records. After the required data cleaning process (data verification, missing values removal in case of some missing attributes, verification of balancing among cars, traffic diversity, etc.), the data set length was 21880 records. It was based on journeys made by a number of various cars.

During the test drives, we focused on providing a reasonable variance of the data. The planned drives were conducted in both day and night conditions, during different seasons, and by a number of different drivers. We drove in and out of built-up areas, on the highway and in heavy traffic. We also used several electric vehicles (Hyundai Ioniq, Nissan Leaf, Skoda CitiGo, Tesla Model 3, Tesla Model S, Volkswagen ID 3).

We included 1-second sampled readouts of a range of attributes such as geographical position, altitude, road slope, traffic intensity, vehicle speed, length of the distance traveled, weather conditions (temperature, wind direction and speed), as well as the state of battery charge, energy gain or loss on a given section of the road, vehicle mileage, and an indication of activated systems (for air conditioning and heating systems, among others).

Some of these attributes, after an introductory investigation, became the model features. These were the traveled distance, the temperature, the date and time of the trip, the speed limit and the road class, the car and the driver

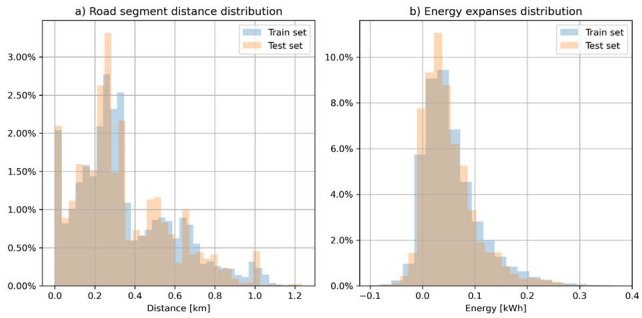

FIGURE 4. The actual road segments (a) and the energy expenditure or gains (b) measurements distribution for training and test datasets, which we evaluated to confirm if the test set supports a fair verification for the trained ML models.

indication, the altitude gained and lost, as well as the maximum acceleration and the speed of the car for the referenced road segment. A preview of the data allocation between the training and test data subsets is shown in Figure 4 where it can be seen that subsets are evenly distributed with respect to energy and distance measured. We used 80% of the collected data to train the algorithms, and the remaining 20% to evaluate their performance. In order to provide a fair test for the algorithms, we stratify the dataset by energy expenditures.

The evaluation of the developed predictors was based on the standard cross-validation procedure, as it is effective in overfitting prevention and allows the verification of the model's performance efficiently [43], [47]. The main objective of the final model was to achieve the lowest possible prediction error. To assess the obtained estimates, we used common regression metrics such as mean absolute error (*MAE*), mean squared error (*MSE*), mean average percentage error (*MAPE*), and symmetric mean absolute percentage error (*sMAPE*) as an improved relative performance indicator [48], and to allow comparability with some other implementations also the coefficient of determination (R^2) as it is still reported in some publications for similar predictions [42]. As the road segments used in the test procedure vary in terms of length, interpretation of the MAE or MSE metric is not necessarily straightforward, so we employed an additional metric – MAE^{100}, which, after simple rescaling, reflects the mean error for a 100 km long test. For n test measurements m_i, the same number of predictions p_i and distances d_i for the test road sections, the metric is computed according to the following formula:

$$MAE^{100} = \frac{\sum_i^n |r_i - p_i|}{\sum_i^n d_i} \cdot 100 \tag{1}$$

This metric should provide some additional perspective and a better understanding of the performance of the tested models. For instance, for the random-forest-based model, we get MAE^{100}=0.979 [kWh / 100 km], and this should be read as the level of mean model error for energy consumed for an average 100 km trip.

As suggested in [47], for such models, increasing the variance of the input data could lead to higher model

TABLE 1. Performance measures for the tested energy consumption prediction with ML models: XGB, FCNN, linear model with L1 or L2 regularisation, RF, ET and SVR. The best results for each metric boldfaced.

Model architecture	R^2 [–]	MAE [kWh]	MAE^{100} [kWh/km]	$sMAPE$ [%]	Inference time [ms]
XGB	**0.995**	**0.231**	**0.606**	**7.943**	95.9
FCNN	0.985	0.374	0.979	12.305	129.0
L_2	0.979	0.451	1.182	12.690	**58.9**
RF	0.974	0.508	1.331	13.800	264.0
ET	0.958	0.609	1.594	14.032	117.0
SVR	0.965	0.668	1.750	16.863	59.3
L_1	0.923	0.846	2.216	19.251	60.0

performance, so we implemented it for the training part of the dataset. Thus, for the measured road segments, we applied manifold bootstrapping with varying road segment lengths to obtain more input samples of a wider range of lengths.

Besides verifying the popular quantitative model metrics, we wanted to get a fast-running predictor that would support quick inference with no significant delays for the main application. Therefore, we also verified the computation times for the model inference process (Table 1), which have been measured separately for each tested model configuration and averaged over a thousand separate runs.

After evaluating dozens of various hyperparameter configurations for the best-evaluated model, the generated predictions showed a decent and satisfactory performance. The test predictions were computed using the resulting XGB model, with a learning rate of $\eta = 0.3$, maximum depth parameter at 6, and the minimum loss reduction $\gamma = 0$. This model has a mean average percentage error (checked using both $MAPE$ and $sMAPE$) of less than 10%, and the absolute error for the relative 100 km drive test of much less than 1 [kWh] ($MAE^{100} = 0.606$[kWh / 100 km]). It also outperforms other model architectures tested in terms of estimation performance.

This model also allows for sufficiently fast predictions, as it returns predictions in about 96 [ms]. The proposed method is designed to run on the server side, but still, since there could possibly be dozens of client requests, it should not add much overhead to the computation time. The average computation time for this model, as well as for all other models, was measured for their inference process and was benchmarked using a regular desktop PC (on an 8-core AMD®Ryzen 7 3700x processor with 32 GB of RAM). It is also possible to deploy the tested model on at least one class-higher runtime cloud configuration, which should potentially reduce the computation time. The complete list of results for all tested models is shown in Table 1.

We tested the resulting machine-learning models on the unseen data registered during actual test drives. The predictions generated by the model, as well as the residual values for the same set of test data, are presented in Figure 5. One could notice relatively small prediction errors. Moreover, the distribution of these errors is somewhat random and does not indicate any systematic bias.

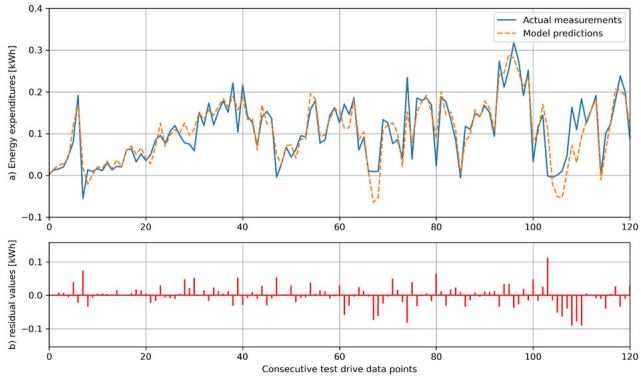

FIGURE 5. A preview of the final machine learning prediction model for the tested road segment: (a) predicted energy expenditures (blue line) compared with measurements (orange dashed line) and (b) the residual values (red ticks) for the following data points of the same road segment.

V. OPTIMIZATION MODEL

The mathematical optimization model provides the solution to the problem of charging an electric car on a predetermined route consisting of a start node, an end node, a number of intermediate nodes and edges between them. It is assumed that on the route or at a certain distance from it, there are charging stations with specific parameters: technical (charging power, type of socket, etc.), economic (charging cost) and other (proximity to an ATM, restaurant, shop, etc.). A route is a directed graph in which each node and edge is described by some parameters, such as edge length, time, time of day, driver habits, driver-dependent travel duration and energy consumption while traversing the edge, etc.

The mathematical model, after being supplied with input data (appropriate for the car, route and driver's preferences), is subject to multi-criteria optimization, which gives the optimal charging schedule for a given car on a given route.

The model has been parameterized in terms of:

- selection of a scalarization function - transforming a multi-criteria task into a single-criterion task,
- providing weights, reference points to specific criteria,
- setting appropriate charging function (linear, piecewise),
- using time slots or not.

An acyclic direct multigraph $G = (N, E)$ was chosen to model the problem (see Figure 6). Nodes $N = \{1, 2, \ldots, |N|\}$ are ordered locations on a selected route, and edges are possible sub-routes between two consecutive nodes N_i, N_{i+1}. The subgraph between two consecutive nodes will be referred to as a segment, and denoted by $s \in \mathcal{S} = \{1, 2, \ldots, |N| - 1\}$. In the segment, there is exactly one edge that does not contain the charging station. The remaining edges of the segment contain exactly one charging station. The route includes $|N|$ nodes, $|E|$ edges and $|S| = |N| - 1$ segments. The number of charging stations equals $|E| - |N| + 1$.

A. TRAVEL AND ENERGY CONSTRAINTS

The journey traverses a sequence of segments. The travel in the segment s begins at the node n_s and ends at the node n_{s+1}.

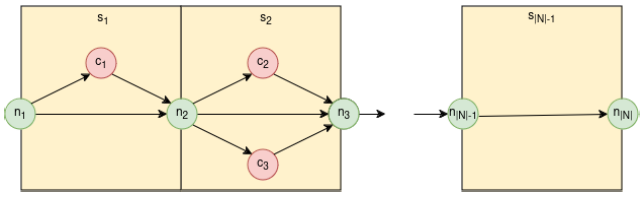

FIGURE 6. The multigraph model of the problem consists of nodes (green circles), edges (black arrows), and charging stations (red circles), grouped into segments.

The time of departure from n_s and time of arrival to n_{s+1} will be denoted by continuous variables tbs_s and tes_s, respectively. The travel within the segment can be done either directly without visiting any of the charging stations, or it can go through one of the charging stations c belonging to the given segment s, i.e. $c \in \mathcal{C}_s$. This decision is represented by the binary variable vcs_s. The travel without visiting any charging stations is modeled with the following time constraint:

$$\forall_{s\in\mathcal{S}} \quad tbs_s + \mathrm{DS}_s * (1 - vcs_s) \leq tes_s, \tag{2}$$

where the DS_s parameter is the travel duration.

The journey along one of the edges with the charging station can be divided into three parts: the journey from the node p_s to the charging station $c \in \mathcal{C}_s$, the charging including waiting for the charger and the journey from the charging station to the node s_s. The respective time constraints are stated as follows:

$$\forall_{s\in\mathcal{S}}\forall_{c\in\mathcal{C}_s} \quad tbs_s + \mathrm{DTCS}_c * vcs_c \leq tbc_c \tag{3}$$

$$\forall_{c\in\mathcal{C}} \quad tbc_c + \mathrm{DW}_c * vcs_c + tc_c \leq tec_c \tag{4}$$

$$\forall_{s\in\mathcal{S}}\forall_{c\in\mathcal{C}_s} \quad tec_c + \mathrm{DFCS}_c * vcs_c \leq tes_s \tag{5}$$

where DTCS_c and DFCS_c are respective travel duration parameters, tbc_c and tec_c are variables expressing charging start and end times, DW_c is the waiting duration parameter, tc_c is the decision variable expressing the duration of the charging process (see section V-C) and vcs_c binary variable is set to 1 if the particular charger c is utilized.

The logical decision to charge (or not) in each segment s is strictly linked to the decision of which specific charger $c \in \mathcal{C}_s$ should be used:

$$\forall_{s\in\mathcal{S}} \quad \sum_{c\in\mathcal{C}_s} vcs_c = vcs_s \tag{6}$$

There are also time constraints put on consecutive segments:

$$\forall_{s\in\mathcal{S}/s_1} \quad tes_{s-1} \leq tbs_s \tag{7}$$

where s_1 is the first segment of the route, tes_{s-1} and tbs_s are the continuous variables denoting the time of leaving segment $s-1$ and the time of entering segment s, respectively.

Another aspect of driving an electric car on a given route is energy consumption. The model is based on the conservation of energy rule. This means that the total energy consumed while traveling must be equal to the energy charged. Of course, the initial and the remaining energy stored in the battery (denoted by bat_{n_s} and $bat_{n_{s+1}}$, respectively) and the battery capacity $\overline{\mathrm{A}}$ also need to be considered.

The equation of energy conservation for each segment $s \in \mathcal{S}$, together with the battery maximum capacity constraint, can be stated as follows:

$$\forall_{s\in\mathcal{S}} \quad \sum_{c\in\mathcal{C}_s} cs_c - \mathrm{L}_s * (1 - vcs_s) - \sum_{c\in\mathcal{C}_s} \mathrm{LTC}_c * vcs_c - \\ - \sum_{c\in\mathcal{C}_s} \mathrm{LFC}_c * vcs_c = bat_{n_{s+1}} - bat_{n_s} \tag{8}$$

$$\forall_{s\in\mathcal{S}}\forall_{c\in\mathcal{C}_s} \quad cs_c \leq \overline{\mathrm{A}} * vcs_c \tag{9}$$

where L_s parameter is the energy used on the edge without the charging station, LTC_c and LFC_c parameters denote the energy consumed on the edge to the charging station and from the charging station, respectively, and cs_c variable is the energy charged at the charging station $c \in \mathcal{C}_s$.

B. TIME SLOTS

The traffic and weather conditions, and thus the travel times, energy consumption and related costs, undergo heavy fluctuations over the day. We model those fluctuations by dividing the 24-hour horizon into a number of time slots with constant travel parameters. The number of time slots can be as low as 4, corresponding to morning, midday, afternoon and night traffic, or much higher, like 24, representing each hour of the day. In general, all slots must cover the entire period of travel and must not have common parts. However, the number of binary variables needed to model the time slots equals the Cartesian product of the time slots and the parameters expressing any travel times or waiting times on the route. Keeping the number of time slots low helps maintain high model performance. On the other hand, using a larger number of time slots, adjusted for the actual differences between travel times at specific times, allows for greater accuracy, but requires the use of an accurate model and, consequently, longer computational time. Since the business requirement of the model is a fast response, the analysis of the model computation times in relation to the applied parameterization is presented in the further part of the paper.

In the following example, the time slots approach is used to model the travel along a single edge without any charging stations. For each segment $s \in \mathcal{S}$ and for each time slot $z \in \mathcal{Z}$, the following travel parameters of an edge without a charging station are given: $\mathrm{DS}_{s,z}$ travel time, BEG_z slot starting time, and END_z slot ending time.

Also, for the edge without charging station in segment $s \in \mathcal{S}$ and time slot $z \in \mathcal{Z}$, the following variables are used: tbs_s – the moment of starting the travel, tes_s – the moment of ending the travel and $vs_{s,z}$ – binary variable equals 1 if the vehicle starts the travel along the edge in the time zone z.

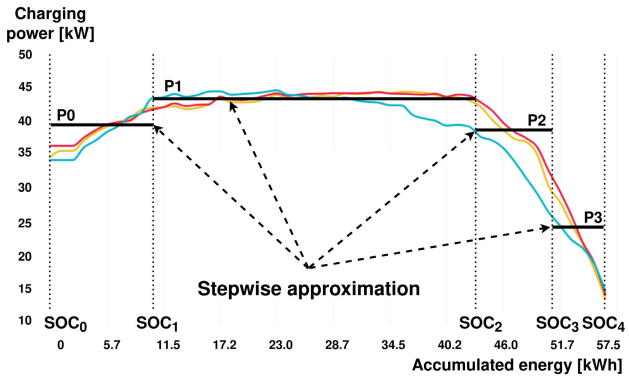

FIGURE 7. Approximation of the Tesla Model 3 charging curve (20°C external temperature – red). Other temperatures: -20°C – blue, 0°C – yellow (source: [49]).

The final model can be stated as follows:

$$\forall_{s\in\mathcal{S}} \quad tbs_s + \sum_{z\in\mathcal{Z}} \mathrm{DS}_{s,z} * vs_{s,z} \leq tes_s \tag{10}$$

$$\forall_{s\in\mathcal{S},z\in\mathcal{Z}} \quad \mathrm{BEG}_z * vs_{s,z} \leq tbs_s \leq \mathrm{END}_z + M * (1 - vs_{s,z}), \tag{11}$$

where M is a sufficiently big constant.

The constraint (10) binds the travel starting time, the travel duration in different time slots and the travel ending time. The proper time slot is selected in the equation (11).

C. CHARGING TIMES

The charging curve is modeled as a step function with consecutive steps P_0, P_1, P_2, and P_3, which approximate the corresponding segments of the original function. These segments are separated by the accumulated State of Charge values: SOC_1, SOC_2, and SOC_3. An example approximation of the Tesla Model 3 charging curve, using a CHAdeMO charging adapter, is shown in Figure 7. The figure depicts three charging curves for different external temperatures: −20°C (blue), 0°C (yellow), and 20°C (red). The stepwise approximation is applied to the red curve.

The aim of the model is to compute the charging time. This, however, depends on two variables: the battery charge state at the charging station c just before charging (accumulated energy) bat_{pcs_c} and the energy cs_c added at the charging station c. Thus, the charging time cannot be modeled as a simple piecewise linear function of one variable.

The proposed approximation model requires mild assumptions to be made. The first assumption applies to the general shape of the charging curve or, more specifically, to the approximating steps, which should meet the following constraints: $\mathrm{P}_0 \leq \mathrm{P}_1 \geq \mathrm{P}_2 \geq \mathrm{P}_3$. The second assumption states that the values SOC_1 and SOC_3, describe approx. 20% and 80% of the battery capacity. The last assumption states that the charging time tc_c at the charge station c is to be minimized.

With the above assumptions, the charging time approximation model, which uses just two binary variables ${}^{H}ba_c$ and ${}^{L}bac_c$ per charge station can be stated as follows:

$$\forall_{c\in\mathcal{C}} \quad bat_{pcs_c} \geq {}^{H}a_c + Ha_c \tag{12}$$

$$\forall_{c\in\mathcal{C}} \quad bat_{pcs_c} + cs_c \geq {}^{L}ac_c + Lac_c \tag{13}$$

$$\forall_{c\in\mathcal{C}} \quad SOC_2 * {}^{H}ba_c \leq Ha_c \leq SOC_2 \tag{14}$$

$$\forall_{c\in\mathcal{C}} \quad {}^{H}a_c \leq M * {}^{H}ba_c \tag{15}$$

$$\forall_{c\in\mathcal{C}} \quad SOC_1 * {}^{H}ba_c \leq Lac_c \leq SOC_1 \tag{16}$$

$$\forall_{c\in\mathcal{C}} \quad {}^{L}ac_c \leq M * {}^{L}bac_c \tag{17}$$

$$\forall_{c\in\mathcal{C}} \quad Lac_c - bat_{pcs_c} \leq ene_1 \tag{18}$$

$$\forall_{c\in\mathcal{C}} \quad bat_{pcs_c} + cs_c - SOC_2 \leq ene_2 \tag{19}$$

$$\forall_{c\in\mathcal{C}} \quad bat_{pcs_c} + cs_c - SOC_3 \leq ene_3 \tag{20}$$

$$\forall_{c\in\mathcal{C}} \quad tc_c = (1/\mathrm{P}_0 - 1/\mathrm{P}_1) * ene_1 + 1/\mathrm{P}_1 * cs_c + (1/\mathrm{P}_2 - 1/\mathrm{P}_1) * (ene_2 - {}^{H}a_c) + (1/\mathrm{P}_3 - 1/\mathrm{P}_2) * ene_3 \tag{21}$$

In the model, the constraints (12), (14) and (15), are responsible for splitting the initial battery state bat_{pcs_c} into values below SOC_2 and above SOC_2, (variables ${}^{H}a_c$ and Ha_c, respectively).

Following constraints (13), and (16), are responsible for splitting the final battery state (after charging) into values below SOC_1 and above SOC_1 (variables ${}^{L}ac_c$ and Lac_c).

The next constraint (18) ensures the variable ene_1 represents only the energy taken (during charging) up to the level SOC_1 (the non-negative difference, between SOC_1 and initial charge).

Another constraints (19) and (20), ensure the variables ene_2 and ene_3 represent the non-negative difference between the final state of the battery and the levels SOC_2 and SOC_3, respectively.

Finally, the constraint (21), computes the charging time.

The above model of the stepwise charging function introduces some inaccuracy (in addition to the stepwise function itself). The reason was to have just 2 instead of 4 binary variables. The inaccuracy shows up only when $bat_{pcs_c} > SOC_3$, which happens only when the car with almost full battery tries to charge. In such a case, the model returns slightly longer charging times (is conservative with respect to the total travel time).

D. ERROR OF TIME APPROXIMATION

The piecewise linearization method is a widely used technique for simplifying complex, nonlinear problems into a form that enables efficient solutions using linear programming (LP) or mixed-integer linear programming (MILP). When properly designed, it yields solutions that are nearly optimal. However, this approach involves a trade-off in accuracy, and the quality of the results depends on the linearization method employed [50]. In some cases, it may lead to suboptimal solutions (optimum distortion) that do not correspond to the global optimum of the nonlinear function [51]. The primary advantage of linearization is the ability to create more efficient, large-scale LP or MILP

models that can be solved quickly and yield optimal solutions within the model space.

In our approach, we employed a coarser stepwise approximation of the battery charging curve. Due to space constraints, this article does not elaborate on the construction of the stepwise approximation. However, under mild assumptions, we provide a comprehensive error analysis for this approximation in the context of charging time estimation.

Let $P(SOC)$ represent the charging curve, which expresses the charging power as a function of the State of Charge. The exact time to charge the battery from SOC_a to SOC_b is given by the integral:

$$T_{a,b} = \int_{SOC_a}^{SOC_b} \frac{1}{P(SOC)}, dSOC \tag{22}$$

In the proposed stepwise approximation, the charge range is divided into intervals defined by points $SOC_0, SOC_1, \ldots, SOC_4$, where SOC_0 and SOC_4 are the minimum and the maximum possible charge levels, respectively. For each interval $i = 1, \ldots, 4$, with bounds SOC_{i-1} and SOC_i, $SOC_{i-1} < SOC_i$, the charging curve is approximated by a constant power P^i.

For simplicity, let us assume SOC_a lies in interval $[SOC_{k-1}, SOC_k]$ and SOC_b lies in interval $[SOC_{l-1}, SOC_l]$ with $k \leq l$. The approximate time of charging from SOC_a to SOC_b over multiple intervals is:

$$T_{a,b}^{\text{approx}} = \frac{SOC_k - SOC_a}{P^k} + \sum_{i=k+1}^{l-1} \frac{SOC_i - SOC_{i-1}}{P^i} + \frac{SOC_b - SOC_{l-1}}{P^l} \tag{23}$$

If SOC_a and SOC_b lie within the same interval i with a constant power approximation P^i, the charging time simplifies to:

$$T_{a,b}^{\text{approx}} = T_{a,b}^{\text{const,i}} = \frac{SOC_b - SOC_a}{P^i} \tag{24}$$

The optimization model employs this stepwise approximation to estimate vehicle charging time. However, the approximation introduces an error in the time estimation, computed as:

$$\mathrm{E}_{a,b} = T_{a,b}^{\text{approx}} - T_{a,b}, \quad \mathrm{RE}_{a,b} = \frac{T_{a,b}^{\text{approx}}}{T_{a,b}} - 1, \tag{25}$$

where Ea, b is the absolute error, and RE$a, b$ is the relative error.

1) HARMONIC MEAN APPROXIMATION

To minimize the time-approximation error, the constant power for interval i can be set to the harmonic mean of $P(SOC)$ over the interval. The harmonic mean, denoted by $\widetilde{P}^i$ ensures that the approximate charging time equals the exact time when charging over the entire interval $[SOC_{i-1}, SOC_i]$. From (22) and (23):

$$\frac{SOC_i - SOC_{i-1}}{\widetilde{P}^i} = \int_{SOC_{i-1}}^{SOC_i} \frac{1}{P(x)} dx$$

Thus:

$$\widetilde{P}^i = \left(\frac{1}{SOC_i - SOC_{i-1}} \int_{SOC_{i-1}}^{SOC_i} \frac{1}{P(x)} dx \right)^{-1} \tag{26}$$

By definition, no time-approximation error occurs when the charging range matches the full interval. Errors arise when the initial or final SOC lies within an interval, as the constant $\widetilde{P}^i$ may not accurately reflect $P(SOC)$ over a partial interval.

2) WORST-CASE ERROR ANALYSIS

The maximum time-approximation error ($\mathrm{E}_{\max}^i$) or its relative variant ($\mathrm{RE}_{\max}^i$) occurs when the difference between the approximate power $\widetilde{P}^i$ and the actual power $P(SOC)$ is largest within interval i. Without loss of generality, consider the case where the actual power is lower than the approximate power, leading to an underestimation of the charging time, which should be avoided, for example in risk-averse solutions. Let $P_{\min}^i = \min_{SOC \in [SOC_{i-1}, SOC_i]} P(SOC)$. For a short charge where the actual power is approximately $P_{\min}^i$, the time approximation error is expressed as:

$$\mathrm{E}_{\max}^i = \frac{1}{P_{\min}^i} - \frac{1}{\widetilde{P}^i}, \quad \mathrm{RE}_{\max}^i = \frac{P_{\min}^i}{\widetilde{P}^i} - 1 \tag{27}$$

for its absolute and relative variants.

This error can be significant if $P_{\min}^i$ is much smaller than $\widetilde{P}^i$. For typical charging curves, the error depends on the interval size and the steepness of $P(SOC)$. However, charging small amounts of energy is uncommon due to significant setup times and costs, such as travel to charging stations and waiting for a charger. Larger charge amounts within the same interval typically result in a smaller error, as the harmonic mean better represents the average behavior over the interval.

3) LINEAR POWER FUNCTION, SINGLE INTERVAL CHARGING

Consider an interval i and $P^i(SOC)$ as a linear approximation of the charging curve in this interval. Let $\bar{P}_x^i = P^i(SOC_x)$, for some charge level SOC_x in interval i. Furthermore, let $\bar{P}_{i-1}^i$ and $\bar{P}_i^i$ be the values of $P^i(SOC)$ at the beginning and at the end of interval i, respectively. Then $P^i(SOC)$ can be expressed as linear function of SOC $P^i(SOC) = \bar{P}_{i-1}^i + m^i(SOC - SOC_{i-1})$, where $m^i = \frac{\bar{P}_i^i - \bar{P}_{i-1}^i}{SOC_i - SOC_{i-1}}$ and the exact charging time within the interval i can be calculated as:

$$T_{a,b}^{\text{lin},i} = \int_{SOC_a}^{SOC_b} \frac{dx}{P^i(x)} = \frac{1}{m^i} \ln\left(\frac{\bar{P}_b^i}{\bar{P}_a^i} \right), \tag{28}$$

where $\bar{P}_b^i \neq \bar{P}_a^i$; otherwise, the basic formula (24) applies.

The linear charging curve approximation is further approximated by the harmonic mean $\widetilde{P}^i$, which in the linear case can be derived from (26) as logarithmic mean of $\bar{P}_{i-1}^i$ and $\bar{P}_i^i$:

$$\widetilde{P}^i = \frac{\bar{P}_i^i - \bar{P}_{i-1}^i}{\ln\left(\frac{\bar{P}_i^i}{\bar{P}_{i-1}^i} \right)}, \text{ for } \bar{P}_i^i \neq \bar{P}_{i-1}^i \tag{29}$$

It can practically useful to estimate the relative time approximation error within a single interval i, starting at the

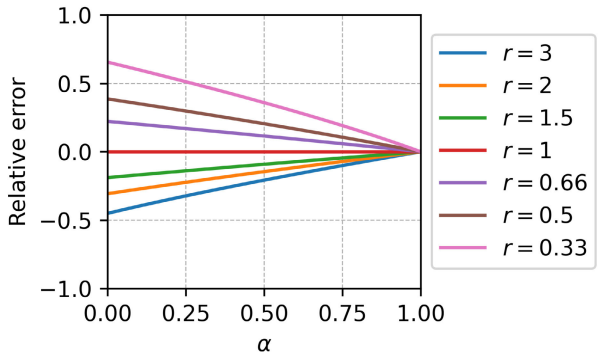

FIGURE 8. Relative time approximation error when charging in the leftmost α-fraction of the single charge interval (i.e., starting at the beginning of the interval and ending at its *alpha*-fraction); *r* is the ratio between the charging power at the end and the charging power at the beginning of the interval *i*.

charge SOC_{i-1} and ends at some charge $SOC_b \leq SOC_i$, or alternatively, starts at some charge $SOC_a \geq SOC_{i-1}$ and ends at SOC_i. Without loss of generality, we discuss the first option and calculate the relative time approximation error $\mathrm{RE}^{\mathrm{lin}}_{i-1,b}$.

Exact charging time in the range $[SOC_{i-1}, SOC_b]$ can be directly expressed with (28) by substituting a for $i-1$:

$$T^{\mathrm{lin}}_{i-1,b} = \int_{SOC_{i-1}}^{SOC_b} \frac{dx}{P^i(x)} = \frac{1}{m^i} \ln\left(\frac{\bar{P}^i_b}{\bar{P}^i_{i-1}}\right) \tag{30}$$

assuming that the power function is not constant. Otherwise, the formula (24) should be used.

The *approximate charging time* for the same SOC range, computed using the stepwise approximation of $P^i(SOC)$ with step value $\widetilde{P}^i$ expressed in (29) is calculated as:

$$T^{\mathrm{approx}}_{i-1,b} = \frac{SOC_b - SOC_{i-1}}{\widetilde{P}^i}$$

With all necessary formulae, we can express the relative time approximation error as:

$$\mathrm{RE}^{\mathrm{lin}}_{i-1,b} = \frac{T^{\mathrm{approx}}_{i-1,b}}{T^{\mathrm{lin}}_{i-1,b}} - 1 = \frac{\alpha \ln r}{\ln(1+\alpha(r-1))} - 1, \tag{31}$$

where $\alpha = \frac{SOC_b - SOC_{i-1}}{SOC_i - SOC_{i-1}}$, $r = \frac{\bar{P}^i_i}{\bar{P}^i_{i-1}}$, and $\bar{P}^i_i \neq \bar{P}^i_{i-1}$, while for $\bar{P}^i_i = \bar{P}^i_{i-1}$, the relative error is 0.

Figure 8 illustrates the relative error of the time approximation based on a stepwise approximation of a linear charge curve in a single interval. More precisely, it shows the relative time approximation error when charging in the leftmost α-fraction of the charge interval.

4) LINEAR POWER FUNCTION, MULTIPLE INTERVALS

When charging across multiple intervals, from SOC_a in interval $[SOC_{k-1}, SOC_k]$ to SOC_b in $[SOC_{m-1}, SOC_m]$ ($k \leq m$), the total approximated time $T^{\mathrm{approx}}_{a,b}$ is the sum of contributions from partial intervals at the start and end, and full intervals in between, with constant power approximation as the harmonic mean $\widetilde{P}^i$ in the respective interval i; see (23), (24), and (29):

$$T^{\mathrm{approx}}_{a,b} = T^{\mathrm{approx}}_{a,k} + \sum_{i=k+1}^{m-1} T^{\mathrm{approx}}_{i-1,i} + T^{\mathrm{approx}}_{m-1,b} \tag{32}$$

The exact time is calculated based on the general formula (22), which, for a piecewise linear power function approximation, breaks into the analytical formula (28) for $T^{\mathrm{lin},i}_{a,b}$, defined separately for each interval i:

$$T_{a,b} = T^{\mathrm{lin}}_{a,k} + \sum_{i=k+1}^{m-1} T^{\mathrm{lin}}_{i-1,i} + T^{\mathrm{lin}}_{m-1,b} \tag{33}$$

The total error can be readily calculated using equations (25).

Although the worst-case error can be arbitrarily large and is bounded only by the maximum difference between actual power and its approximation (see (27)), it becomes much smaller with more energy charged, due to the averaging effect over multiple intervals. The use of a harmonic mean for the stepwise approximation further minimizes the time approximation error, resulting in no error for full-interval charging. For partial intervals where $P(SOC)$ is approximated by a linear function, this error can be precisely quantified.

For charges spanning multiple intervals, the contribution from the outermost partial intervals plays a crucial role. Designing the stepwise approximation while keeping the number of intervals low requires careful planning and possible interval optimization.

E. MULTIPLE CRITERIA

One of the distinguishing features of the approach is the simultaneous optimization of multiple criteria relevant to the end user. These criteria include travel time, charging costs, number of charging events, risk assessment of prolonged charges and, last but not least, driver's satisfaction regarding the desired additional amenities at selected charging stations. A brief discussion of these criteria and the aggregation method is presented below.

The travel time k^{mt} is simply calculated as the difference between the end of the last segment and the beginning of the first segment of the route. This includes the driving time, the charging station waiting time and the charging time itself.

The charging cost criterion (variable k^{tc}) is calculated as the total of the energy charged on all charging stations multiplied by the energy unit cost for each station. To prevent the model from leaving the empty battery at the travel end, an additional cost is applied that accounts for the loss of the accumulated energy.

We have also included the number of charging events (variable k^{nc}) as one of the optimisation criteria, to allow drivers (and passengers) to express their preferences regarding the number of stops to charge the vehicle during the journey.

One of the essential analyses required when planning any activity, particularly electric vehicle travel, is risk assessment.

There is a wide literature on optimization under uncertainty (see e.g. [52], [53] for generic problems, and [54] for notes of routing under uncertainty). In the considered problem, however, the route is already chosen, so the only sources of uncertainty lie in the travel times on the selected edges, the charger waiting times, and the charging times. The results of our initial real-live experiments show that the charger waiting times have the greatest impact on the uncertainty of the travel time of the entire route.

Our approach takes as one of the criteria the mean value of a specified fraction (10% in this case) of the longest waiting and charging times recorded historically for the specific charging stations. The criterion doesn't pretend to estimate the variability distribution of the waiting and charging times. Rather, it is a simple means of avoiding some of the charging stations with long worst case waiting times, depending on the driver's preferences. The criterion (variable k^r) can be expressed as follows:

$$k^r = \sum_{c \in \mathcal{C}} vcs_c * \mathrm{DW}_c^{10\%} + \sum_{c \in \mathcal{C}} tc_c * \mathrm{TC}_c^{\Delta} \tag{34}$$

where vcs_c is binary variable set to 1 if the charging station c is used, and $\mathrm{DW}_c^{10\%}$ parameter expresses average of the 10% longest waiting times for charger c. The tc_c continuous variable denotes the charging time and TC_c^{Δ} is the average elongation from the 10% longest charges at the charging station c as compared to the mean of all charging times.

The last criterion expresses the driver's dissatisfaction regarding desired additional amenities $a \in A$ (including a restaurant, hotel, ATM, and a store) at the visited charging stations $vcs_c = 1$. The criterion was formulated as the sum over charging station $c \in \mathcal{C}$ of vcs_c variables times weighted with user preferences PREF^a to each amenity:

$$k^{ame} = \sum_{a \in \mathcal{A}} \mathrm{PREF}^a * \sum_{c \in \mathcal{C}} (vcs_c * (1 - has_{c,a})) \tag{35}$$

Certainly, other amenity aggregations are possible, the total number of amenities of a certain type on the whole trip being a good example. In addition, different constraints can be introduced to express hard requirements regarding the presence of some amenities.

F. CRITERIA AGGREGATION

In order to make an optimization model operational, one needs to assume some solution concept specifying what it means to optimize multiple objective functions. In decision support systems, the aggregation function must be constructed in such a way that the decision maker's preferences regarding the importance of the criteria can be specified. Two such approaches are commonly used for such problems: the weighted sum and the reference point. Weighted sum aggregation, when properly scaled, is easily understood by the end-user and can be embedded in a UX interface. However, in the case of non-linear problems, especially MIP, it has the property of not being able to access every non-dominated solution from the criterion space, i.e., not every Pareto optimal solution from the decision space, regardless of the weights provided by the end user. However, aggregation of reference points can generate any Pareto optimal solution when provided with appropriate reference points that express the preferences of the end user (see [55] for a detailed description). This comes at the cost of a more demanding modeling and a less user-friendly process to reach the final decision.

Instead of using a full-featured reference point method, a simplified version is sometimes used, which takes the form of a minimax scalarisation. It is similar in terms of the general decision process and allows one to reach all non-dominated solutions. The disadvantage is that in some degenerate cases, when using classical LP or MIP solvers, there is no guarantee that a non-dominated solution will always be obtained.

All three scalarization methods, weighted sum, reference point, and minimax, were made available for selection by the end user in the final version of the model.

For the purposes of the computational experiments, however, we decided to use weighted sum aggregation, with the objective function taking the form of:

$$\min O = \mathrm{w}^{mt} * k^{mt} + \mathrm{w}^{tc} * k^{tc} + \mathrm{w}^{nc} * k^{nc} + \mathrm{w}^{r} * k^{r} + w^{ame} * k^{ame}, \tag{36}$$

where w^{mt}, w^{tc}, w^{nc}, w^{r} and w^{ame} are respective weight parameters applied to each criterion taken into account. It should be noted that the final weighting values of the objective function should be determined by the end user of the application, based on their preferences regarding the planned journey. These preferences may include achieving the shortest travel time, the lowest charging costs, or stopping at a station with the required amenities. The application should enable a comparison of the algorithm outcomes for multiple selected priorities related to the defined travel criteria. Ultimately, the choice of the solution should remain with the end user.

VI. COMPUTATIONAL EXPERIMENTS AND FIELD TESTS

The following experiments were carried out to examine the robustness, reliability, and effectiveness of the optimization model, as well as its performance.

A. PERFORMANCE ANALYSIS

The analysis of the performance of the mixed-integer linear mathematical model described in this paper was carried out using a long and complex travel route from Świnoujście (a city in the northwest of Poland) to Przemyśl (a city in the southeast part of Poland) with the total length of the route of over 984 km. The route comprises 68 nodes, 67 edges, 62 charging stations, and 23 time slots. The large number of time slots was chosen to create a worst-case scenario for comparison purposes. In real-world conditions, periods with similar travel parameters, such as nighttime, can often be consolidated into a single time slot. The calculations were performed on a computing server equipped

TABLE 2. Model statistics for EV charging time func. approach.

	Model with Time Zones		Model without Time Zones (REF)	
	SL	CP	SL	SP
Equation blocks	35	47	22	34
Variables blocks	20	29	16	25
Non-zero elements	42,944	44,928	2,401	4,385
Single equations	12,748	13,492	856	1,600
Single variables	6,405	6,963	586	1,144
Discrete variables	5,948	6,072	129	253
Solution time [s]	5.047	29.750	0.088	0.365

with a CPU: Intel Core i7-8086K @4.00 Ghz, 48 GB of RAM, Windows 10 Pro, using GAMS software version 31.2 and CPLEX solver version 12.10.0. Both implemented approaches to calculating the charging time of an electric vehicle were tested in the mathematical model, i.e., the simplified linear (SL) approach and the complex piecewise (CP) approach to the linearization of the non-linear charging time curve. For comparison purposes (as a reference point), calculations for the described travel route were made using a mathematical model without implemented time slots. The model statistics, including a number of single equations, variables, and binary variables, are presented in Table 2. Model statistics. For the analyzed case, considering the complex vehicle charging time estimation function, 13,492 equations, 6,963 variables, and 6,072 binary variables were generated (for the simplified vehicle charging approach, the values were correspondingly lower). The version of the model without time slots was characterized by much lower values of the generated model elements, which amounted to 1,600 single equations, 1,144 variables, and 253 binary variables, respectively. When analyzing the obtained computation times it should be emphasized that the solver tolerances (CPLEX) were as follows: relative stopping tolerance (epgap) - 1.0e-04, the absolute stopping tolerance (epagap) - 0, the integrity tolerance (epint) - 1.0e-05 and the feasibility tolerance (eprhs) - 1.0e-06. Differences in the level of complexity of assumed configurations of the mathematical model largely influenced the average model solution times (Table 2). Regardless of the vehicle charging time function adopted, the solution time of the reference model (without time zones) was well below 1 second (without time zones: 0.09 [s], with time zones: 0.4 [s]). However, in the case of the model with time zones for the simplified vehicle charging time function, the average solution time was over 5 seconds. For the complex vehicle charging time function, it was almost 30 seconds.

B. CASE STUDY

The presented case study aimed to investigate the optimal, according to assumed objective function scalarization priority, charging station selection (including location and schedule) for electric cars along the developed *R_WZ_TZ* travel route. The analyzed route, which assumes driving from the starting node *A* located in Warsaw to end node *B* located in Zakopane, with a total distance of 412 km, consists of 11 edges, to which 14 charging stations are assigned. The total number of individual chargers is 20 due to multiple charging connectors within a single station. Figure 9 visually represents the route examined in the case study.

TABLE 3. Techno-economic parameters related to the individual chargers and a matrix of additional facilities located near the charging station (1 - facility present; 0 - otherwise).

Charging stations	Parameter		Additional facilities next to the charging station [1:yes , 0:no]			
	Power [W]	Cost [PLN/Wh]	Restau-rant	Hotel	ATM	Store
S1_2_1	100 000	0.00239	1	0	0	0
S1_2_2	50 000	0.00199	0	1	0	1
S2_2	22 000	0.00109	0	1	0	0
S3_2_1	50 000	0.00199	0	0	0	0
S3_2_2	22 000	0.00143	0	1	0	0
S4_2	11 000	0.00120	0	0	0	0
S5_2	22 000	0.00130	0	0	1	0
S6_2	22 000	0.00130	0	0	0	0
S7_2_1	40 000	0.00209	1	1	1	1
S7_2_2	22 000	0.00129	1	0	0	0
S8_2	50 000	0.00199	0	0	0	1
S9_2	50 000	0.00209	0	1	0	1
S10_2_1	50 000	0.00209	0	0	0	0
S10_2_2	22 000	0.00129	1	0	1	0
S11_2_1	100 000	0.00239	0	0	0	0
S11_2_2	50 000	0.00199	0	0	0	0
S12_2	50 000	0.00229	1	1	1	1
S13_2	50 000	0.00199	0	0	0	0
S14_2_1	22 000	0.00099	0	0	0	1
S14_2_2	50 000	0.00145	0	1	0	0

The *R_WZ_TZ* route analyzed in the model includes the following:

- 12 route nodes (nodes), including start node A: N52°13.94195', E020°59.77709' and end node B: N49°17.47329', E019°57.65912', which define where the route begins and ends;
- 11 edges / segments (edges);
- 14 charging stations / 20 charging points (chargeStations).

For each of the analyzed charging stations located directly on the *R_WZ_TZ* route or in its immediate vicinity, a set of parameters was determined that could be crucial in the decision-making process for station selection (Table 3). The rated electrical power of the charger [W] was determined for each station. One key factor affecting the selection of a charging station is the unit cost of electricity charging associated with a particular station and charger (usually dependent on the type of individual charger, such as fast DC chargers vs. slow AC chargers). Furthermore, the availability of additional facilities such as restaurants, stores/groceries, ATMs, or hotels at or near the station may also be significant (related to driver preferences). Consequently, Table 3 assigns the facilities mentioned above to each identified charging station on the *R_WZ_TZ* route, with a value of 1 indicating the presence of the facility and 0 otherwise.

For the analyzed journey on the *R_WZ_TZ* route, both lower and upper limits for the installed capacity of vehicle battery were established. It is also essential to specify the

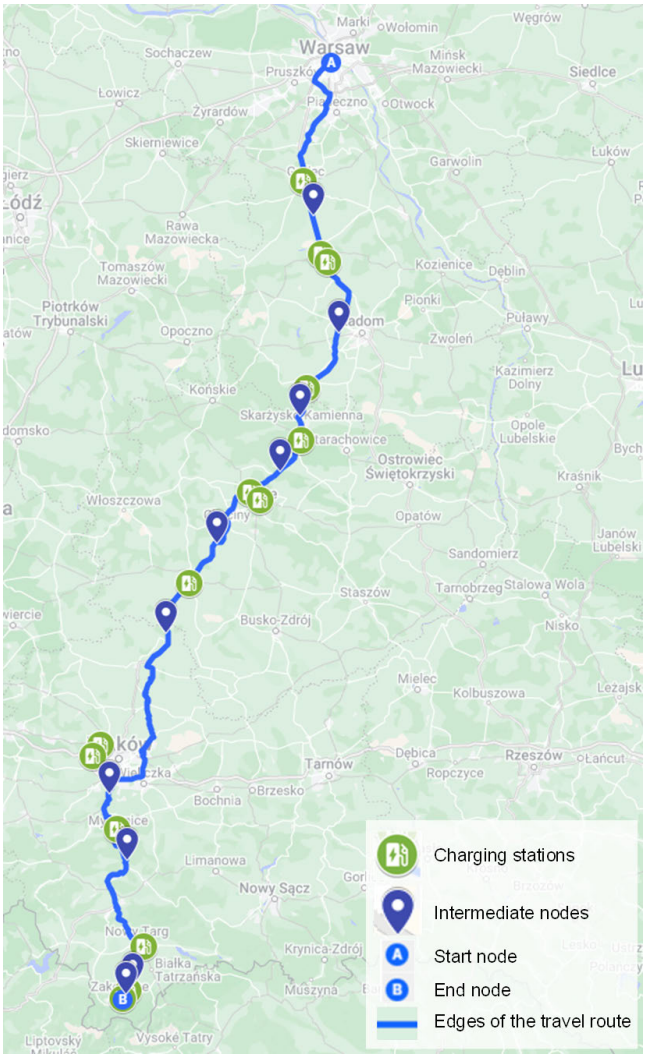

FIGURE 9. *R_WZ_TZ* route – visualization.

TABLE 4. The battery capacity and its state of charge limitations.

Parameter (nodeParam)	Value [Wh]
minTank	800
maxTank	58 000
start_tank_state	30 000
min_final_tank_state	10 000

initial battery state of charge at the beginning of the journey and the required state of charge at the end of the trip, i.e., at the end/final node (Table 4).

In the case study tests, three scenarios were developed. In scenario 1, tests were carried out to lower the total travel time of the journey along the *R_WZ_TZ* route. In scenario 2, the focus was on minimizing the total travel cost of the trip (cost of energy consumption, including cost of charging). In contrast, scenario 3 aimed to minimize the value of penalties incurred for failing to meet driver requirements regarding additional facilities presence next to the charging stations (maximizing the driver preferences fulfillment) (Table 5).

TABLE 5. Case study scenario assumptions regarding the scalarization priority of the optimization model's objective function.

Scenario	Route	Objective function criterion priority
Scenario 1	R_WZ_TZ	Total travel time
Scenario 2	R_WZ_TZ	Total travel cost
Scenario 3	R_WZ_TZ	Driver preferences

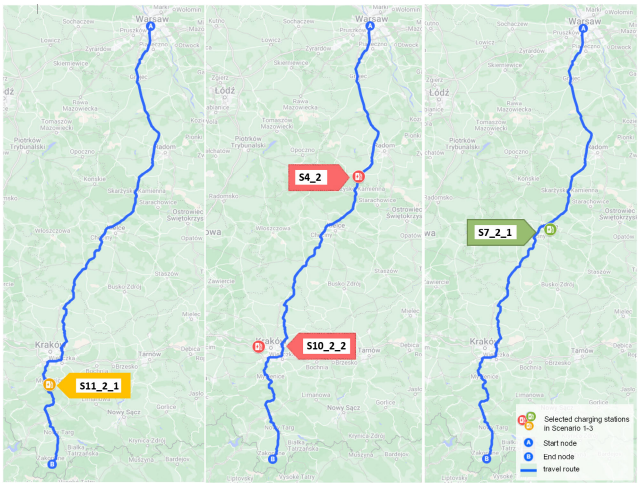

FIGURE 10. Optimization model results - allocated charging stations under analyzed scenarios.

TABLE 6. Optimization results for scenarios 1–3 (*R_WZ_TZ* route).

	Scenario 1 (Time priority)	Scenario 2 (Cost priority)	Scenario 3 (Drivers preferences priority)
	Objective function criteria values		
Total charging cost [PLN]	34.08	**10.09**	82.92
Total travel cost [PLN]	54.08	**30.09**	102.92
Total travel time [h]	**4.22**	5.60	6.22
No. of penalties [-]	3.00	6.00	**0.00**
Delay Risk [-]	1.98	3.90	1.82
	Objective function value		
Weighted Average Scalarization	533.56	3067.62	195.89

C. RESULTS

Based on the presented assumptions and input data for the *R_WZ_TZ* route, calculations were performed using a mathematical model to optimize the process of charging station selection (location and charging schedule) under assumed scenarios (1)-(3). The outcomes of the model have been visualized in Figure 10.

The summary results for all analyzed scenarios in the case study (*R_WZ_TZ* route) has been presented in Table 6.

D. SCENARIO 1

According to the obtained results and to meet all the user-defined requirements (objective function scalarization in scenario 1 - priority of travel time minimization), station *S*11_2_1 was selected. The selected station and the value of

TABLE 7. Selected charging stations for scenarios 1–3.

Scenario	Charging stations	Electricity charged at the station [Wh]
Scenario 1	S11_2_1	14 258.86
Scenario 2	S4_2	7763.65
	S10_2_2	599.91
Scenario 3	S7_2_1	39674.29

energy charged have been presented in Table 7. The total travel time for route *R_WZ_TZ* in scenario 1 was 4 hours and 13 minutes. The cost of charging (*crit_tank_cost*) amounted to 34.08 PLN. The total travel cost, including the charging cost and the cost of the remaining energy consumed during the journey (*totalCost*), was 54.08 PLN. The results also indicate three penalties for not meeting additional driver requirements (additional facilities at the stations). The risk value associated with a possible extension of the period spent at the station (resulting from uncertainties related to charging or waiting times) is 1.98. The objective function value, including the costs of consumed energy to cover the entire route (including battery charging) and the costs incurred due to not meeting the specified driver requirements, was 533.56. These results have been presented in Table 6.

E. SCENARIO 2

According to the model results, two charging stations were selected in scenario 2 (priority of travel cost minimization): *S*4_2 and *S*10_2_2. The list of selected stations, along with the value of energy charged at each of them, has been presented in Table 7. The total travel time for route *R_WZ_TZ* in scenario 2 was 5 hours and 36 minutes (approx. 33% longer than in scenario 1). However, the travel costs, including the charging cost (10.09 PLN) and total travel cost (30.09 PLN), were respectively 30% and 55% lower than scenario 1 (which was second best). The results also indicate that five penalties were incurred for not meeting driver requirements regarding additional facilities near the charging stations. Considering the costs of consumed energy (including battery charging) and the costs incurred due to not meeting the driver preferences, the objective function value was 3067.62 (Table 6).

F. SCENARIO 3

In scenario 3 (priority of driver preferences maximization/minimization of a number of penalties), again, only one charging station was selected, namely *S*7_2_1 (Table 7).

The total travel time in scenario 3 was the longest among other scenarios - 6 hours and 13 minutes. Also, the charging (82.92 PLN) and total travel (102.92 PLN) costs were the highest. However, according to objective function priority no penalties were incurred for not meeting driver requirements regarding the presence of the additional facilities near the charging stations. The risk value associated with a possible extension of the period spent at the station (resulting from uncertainties related to charging or waiting times) is 1.82. The objective function value, including the costs of electricity used and the penalty costs for not meeting the driver preferences, was 195.89 (Table 6).

TABLE 8. Techno-economic parameters related to the individual chargers and a matrix of additional facilities located near the charging station (1 - facility present; 0 - otherwise) under Scenario A.

Charging stations	Parameter		Additional facilities next to the charging station [1:yes , 0:no]			
	Power [W]	Cost [PLN/Wh]	Restaurant	Hotel	ATM	Store
		Scenario A				
S1_2_1	100 000	0.00239	1	1	1	0
S1_2_2	50 000	0.00199	1	1	1	0
S2_2	22 000	0.00129	0	1	1	1
S3_2_1	50 000	0.00199	0	1	1	1
S3_2_2	22 000	0.00143	0	1	1	1
S4_2	11 000	0.00120	1	1	1	0
S5_2	22 000	0.00130	0	1	1	1
S6_2	22 000	0.00199	1	1	1	1
S7_2_1	40 000	0.00209	0	1	1	1
S7_2_2	100 000	0.00129	0	1	1	1
S8_2	50 000	0.00199	1	1	1	0
S9_2	50 000	0.00209	0	1	1	1
S10_2_1	50 000	0.00209	1	1	1	0
S10_2_2	22 000	0.00129	1	1	1	0
S11_2_1	100 000	0.00239	0	1	1	1
S11_2_2	50 000	0.00199	0	1	1	1
S12_2	50 000	0.00229	1	1	1	0
S13_2	50 000	0.00199	0	1	1	1
S14_2_1	22 000	0.00099	1	1	1	0
S14_2_2	50 000	0.00145	1	1	1	0

G. SCENARIOS A-D

In order to present a detailed analysis of the decision-making process based on the driver's preferences, four additional scenarios were developed (A-D). For each of them, it was assumed that the driver's preferences include all four available additional facilities, namely, a hotel, a restaurant, an ATM, and a store.

In Scenario A, only one station (S6_2) offers all required amenities. It features a low charging power (22,000 W) and a high charging cost (0.00199 PLN/Wh).

In Scenario B, additionally, station S2_2 is also equipped with all required amenities. This station has the same charging power but a lower cost (0.00129 PLN/Wh) compared to station S6_2.

In Scenario C, in addition to stations S2_2 and S6_2, station S7_2_2 also offers all additional facilities. Furthermore, this station features the same charging cost as S2_2 but a higher charging power (100,000 W).

The last case analyzed in this section, Scenario D, returns to the assumption that only station S6_2 offers all the amenities required by drivers. An additional change is a significant increase in the charging cost from 0.00199 to 0.00299 PLN/Wh.

Table 8 summarizes the charging station parameters assumed for Scenario A, including charging power, charging cost and additional amenities available at each station.

TABLE 9. Changes in techno-economic parameters and availability of additional facilities located near the charging station related to the selected chargers under Scenarios B, C and D.

Charging stations	Parameter		Additional facilities next to the charging station [1:yes , 0:no]			
	Power [W]	Cost [PLN/Wh]	Restau-rant	Hotel	ATM	Store
		Scenario B				
S2_2	22 000	0.00129	1	1	1	1
		Scenario C				
S2_2	22 000	0.00129	1	1	1	1
S7_2_2	100 000	0.00129	1	1	1	1
		Scenario D				
S6_2	22 000	0.00299	1	1	1	1

TABLE 10. Optimization results for Scenarios A-D.

Parameter	Scenario A	**Scenario B**	Scenario C	Scenario D
Allocated charging stations	S6_2	S2_2 S6_2	S7_2_2	S7_2_2
Electricity charged at the station [Wh]	50,512.67	37,188.96 14,795.23	48,590.56	48,590.56
Objective function criteria values				
Total charging cost [PLN]	120.52	97.42	82.68	82.68
Total travel time [h]	8.32	9.06	6.34	6.34
No of penalties [-]	0	0	0	1
Objective function (mean weighted scalarization)	230.81	186.40	152.83	252.83

Changes to the assumptions for scenarios B, C and D, relative to scenario A, are presented in Table 9.

The results show a significant impact of the priority given to maximizing driver preferences (minimizing the number of penalties for not meeting driver preferences) on the decisions regarding the selection of charging stations in each scenario. However, if driver preferences can be satisfied with lower cost or shorter charging time, stations offering a better solution are selected, taking into account additional criteria. In Scenario A, despite the availability of lower cost stations equipped with higher power chargers (but lacking the full set of required facilities), station S6_2 was selected as it is the only station in this scenario offering all the required facilities. In scenario B, two stations - S2_2 and S6_2 - were selected as they were the only ones that met the drivers' preferences and, in addition, the S2_2 station offered a lower charging cost (the model could not select only the S2_2 station due to its location early on the route and the maximum battery capacity of the EV limitation). The results obtained allowed to reduce the total travel costs and the objective function value (despite a slight increase in total travel time). In Scenario C, the result is determined by adding all the required facilities to a station with lower charging costs and higher charging power. As expected, this station (S7_2_2) was selected as it offered no penalties and the lowest travel cost and time compared to the results of Scenarios A and B. The final case, Scenario D, describes a situation where a significant increase in charging costs occurred at station S6_2, the only one with all the required facilities. In the result model, another station is allocated despite the need to pay a penalty. In this scenario, the selected station, S7_2_2, offered a lower charging cost and charging time, providing the lowest possible value of the objective function, despite lacking one of the required facilities. The summary of the results for each scenario (A-D) is presented in Table 10.

VII. SUMMARY

The article discusses the possibility of creating a functional personal mobile app to optimally select an electric car charging station along a specific driving route.

The functional requirements of the application are presented. The key challenges of creating such an application are identified, namely: preparation of the required data for the application, estimation of data for specific route sections using ML models, and an optimization model that calculates the charging schedule of the car along the route, taking into account the preferences and habits of the specific driver. The article omits the process of producing a mobile app.

The obtained model results for the developed scenarios confirmed aligned with the assumptions. For the time priority criterion (scenario 1), the model selected a solution that allows the shortest travel time. Similarly, for scenario 2 (cost priority), the selected charging stations ensured the lowest travel costs along the analyzed route. In scenario 3 (priority of satisfying driver preferences), the lowest penalty for not meeting the specified requirements regarding the presence of additional facilities near charging stations was achieved compared to the other scenarios.

Further research is required for the development of the mobile application, particularly concerning the selection of appropriate weighting parameter value ranges, which are essential for the scalarization of the objective function in the optimization model. Users should be able to choose how to select or allocate charging stations along an electric vehicle's planned route according to their individual priorities. The results of Scenarios A–D demonstrate that prioritizing user preferences for additional amenities at charging stations does not necessarily mean disregarding other objective function criteria, such as travel time and cost. When multiple stations meet user preferences, the model selects stations with higher charging power, which reduces travel time and charging costs. Additionally, stations that do not fully meet all user requirements may be selected if the cost or travel time trade-offs favor alternative solutions. In such instances, however, the end user should make the final decision regarding the travel variant chosen.

It is also worth emphasizing that meeting all model constraints and assumed multi-objective function criteria priorities (Scenarios 1-3 and Scenarios A-D) has proved the proper operation of the developed optimization model.

APPENDIX

Abbreviations and their descriptions utilized in the paper:

Abbreviation	Description
CP	complex piecewise
CPLEX	complex linear programming expert
CSLP	charging station location problem
DT	digital twin
EBM	energy-based model
ET	extra trees
EV	electric vehicle
EVSPP	electric vehicle shortest path problem
LP	linear programming
LSTM	long short-term memory
MAE	mean absolute error
MAPE	mean average percentage error
MIP	mixed-integer programming
MILP	mixed-integer linear programming
ML	machine learning
MSE	mean squared error
FCNN	fully connected neural network
RF	random forest
SL	simplified linear
sMAPE	symmetric mean absolute percentage error
SOC	state of charge
SVR	support vector regression
XGB	extreme-gradient boosting

REFERENCES

[1] F. Teng, Z. Ding, Z. Hu, and P. Sarikprueck, "Technical review on advanced approaches for electric vehicle charging demand management—Part I: Applications in electric power market and renewable energy integration," *IEEE Trans. Ind. Appl.*, vol. 56, no. 5, pp. 5684–5694, Sep. 2020.

[2] Z. Yi, B. Chen, X. C. Liu, R. Wei, J. Chen, and Z. Chen, "An agent-based modeling approach for public charging demand estimation and charging station location optimization at urban scale," *Comput., Environ. Urban Syst.*, vol. 101, Apr. 2023, Art. no. 101949. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0198971523000121

[3] T. Ma and O. A. Mohammed, "Optimal charging of plug-in electric vehicles for a car-park infrastructure," *IEEE Trans. Ind. Appl.*, vol. 50, no. 4, pp. 2323–2330, Jul. 2014.

[4] X. Gong, R. Xiong, and C. C. Mi, "Study of the characteristics of battery packs in electric vehicles with parallel-connected lithium-ion battery cells," *IEEE Trans. Ind. Appl.*, vol. 51, no. 2, pp. 1872–1879, Mar. 2015.

[5] A. Babin, N. Rizoug, T. Mesbahi, D. Boscher, Z. Hamdoun, and C. Larouci, "Total cost of ownership improvement of commercial electric vehicles using battery sizing and intelligent charge method," *IEEE Trans. Ind. Appl.*, vol. 54, no. 2, pp. 1691–1700, Mar. 2018.

[6] B. Wang, D. Zhao, P. Dehghanian, Y. Tian, and T. Hong, "Aggregated electric vehicle load modeling in large-scale electric power systems," *IEEE Trans. Ind. Appl.*, vol. 56, no. 5, pp. 5796–5810, Sep. 2020.

[7] P. K. Dost, P. Spichartz, and C. Sourkounis, "Charging behavior of users utilizing battery electric vehicles and extended range electric vehicles within the scope of a field test," *IEEE Trans. Ind. Appl.*, vol. 54, no. 1, pp. 580–590, Jan. 2018.

[8] Y. Cao, "Online routing and charging schedule of electric vehicles with uninterrupted charging rates," *IEEE Access*, vol. 10, pp. 98572–98583, 2022.

[9] E. Elghanam, A. Abdelfatah, M. S. Hassan, and A. H. Osman, "Optimization techniques in electric vehicle charging scheduling, routing and spatio-temporal demand coordination: A systematic review," *IEEE Open J. Veh. Technol.*, vol. 5, pp. 1294–1313, 2024.

[10] H. Li, Z. Li, L. Cao, R. Wang, and M. Ren, "Research on optimization of electric vehicle routing problem with time window," *IEEE Access*, vol. 8, pp. 146707–146718, 2020.

[11] F. V. Cerna, M. Pourakbari-Kasmaei, R. A. Romero, and M. J. Rider, "Optimal delivery scheduling and charging of EVs in the navigation of a city map," *IEEE Trans. Smart Grid*, vol. 9, no. 5, pp. 4815–4827, Sep. 2018.

[12] P. Liu, C. Wang, J. Hu, T. Fu, N. Cheng, N. Zhang, and X. Shen, "Joint route selection and charging discharging scheduling of EVs in V2G energy network," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 10630–10641, Oct. 2020.

[13] Y. Hui, Q. Wang, Z. Su, N. Cheng, R. Chen, and T. H. Luan, "Joint optimization of personalized route planning and global traffic scheduling: A DT-enabled integrated framework," *IEEE Trans. Veh. Technol.*, vol. 73, no. 10, pp. 1–12, Oct. 2024.

[14] J. Barco, A. Guerra, L. Muñoz, and N. Quijano, "Optimal routing and scheduling of charge for electric vehicles: A case study," *Math. Problems Eng.*, vol. 2017, no. 1, Jan. 2017, Art. no. 8509783.

[15] Y.-H. Jia, Y. Mei, and M. Zhang, "A bilevel ant colony optimization algorithm for capacitated electric vehicle routing problem," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10855–10868, Oct. 2022.

[16] P. Ranacher, R. Brunauer, S. C. Van der Spek, and S. Reich, "A model to estimate and interpret the energy-efficiency of movement patterns in urban road traffic," *Comput., Environ. Urban Syst.*, vol. 59, pp. 152–163, Sep. 2016. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0198971516300898

[17] T. Erdelić, T. Carić, M. Erdelić, L. Tišljarić, A. Turković, and N. Jelušić, "Estimating congestion zones and travel time indexes based on the floating car data," *Comput., Environ. Urban Syst.*, vol. 87, May 2021, Art. no. 101604. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0198971521000119

[18] A. Fathollahi, M. Gheisarnejad, J. Boudjadar, M. Homayounzadeh, and M.-H. Khooban, "Optimal design of wireless charging electric buses-based machine learning: A case study of nguyen-dupuis network," *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 8449–8458, Jul. 2023.

[19] S. Yanatma. (2023). *Access to EV Charging Stations in Europe is 'a Significant Concern'. How do Countries Compare*. [Online]. Available: www.euronews.com/next/2023/09/18/access-to-ev-charging-stations-in-europe-is-a-significant-concern-how-do-countries-compare

[20] ChargeUp. (2023). *State of the Industry 2023 Report*. ChargeUp Europe. [Online]. Available: https://www.chargeupeurope.eu/state-of-the-industry

[21] PSPA. (2020). *New Electromobility Barometer 2020/2021*. PSPA. [Online]. Available: https://psnm.org/wp-content/uploads/2020/11/barometr_nowej_mobilnosci_2020_raport_S_1.pdf

[22] J. Chen, X. Huang, S. Tian, Y. Cao, B. Huang, X. Luo, and W. Yu, "Electric vehicle charging schedule considering user's charging selection from economics," *IET Gener., Transmiss. Distrib.*, vol. 13, no. 15, pp. 3388–3396, Aug. 2019.

[23] B. Sun, Z. Huang, X. Tan, and D. H. K. Tsang, "Optimal scheduling for electric vehicle charging with discrete charging levels in distribution grid," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 624–634, Mar. 2018.

[24] W. Yin, J. Ji, T. Wen, and C. Zhang, "Study on orderly charging strategy of EV with load forecasting," *Energy*, vol. 278, Sep. 2023, Art. no. 127818.

[25] W. Yin and J. Ji, "Research on EV charging load forecasting and orderly charging scheduling based on model fusion," *Energy*, vol. 290, Mar. 2024, Art. no. 130126.

[26] Y. Song and X. Hu, "Learning-based demand-supply-coupled charging station location problem for electric vehicle demand management," *Transp. Res. D, Transp. Environ.*, vol. 125, Dec. 2023, Art. no. 103975.

[27] S. Çelik and Ş. Ok, "Electric vehicle charging stations: Model, algorithm, simulation, location, and capacity planning," *Heliyon*, vol. 10, no. 7, Apr. 2024, Art. no. e29153.

[28] Y. Li, J. Wang, W. Wang, C. Liu, and Y. Li, "Dynamic pricing based electric vehicle charging station location strategy using reinforcement learning," *Energy*, vol. 281, Oct. 2023, Art. no. 128284.

[29] A. Fathollahi, S. Y. Derakhshandeh, A. Ghiasian, and M. H. Khooban, "Utilization of dynamic wireless power transfer technology in multi-depot, multi-product delivery supply chain," *Sustain. Energy, Grids Netw.*, vol. 32, Dec. 2022, Art. no. 100836. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352467722001217

[30] M. Dastpak, F. Errico, O. Jabali, and F. Malucelli, "Dynamic routing for the electric vehicle shortest path problem with charging station occupancy information," *Transp. Res. C, Emerg. Technol.*, vol. 158, Jan. 2024, Art. no. 104411.
[31] M. Cubillos, M. Dell'Amico, O. Jabali, F. Malucelli, and E. Tresoldi, "An enhanced path planner for electric vehicles considering user-defined time windows and preferences," *Energies*, vol. 16, no. 10, p. 4173, May 2023.
[32] N. D. Kullman, A. Froger, J. E. Mendoza, and J. C. Goodson, "Frvcpy: An open-source solver for the fixed route vehicle charging problem," *INFORMS J. Comput.*, vol. 33, pp. 1277–1283, Feb. 2021.
[33] S. M. Nourbakhsh and Y. Ouyang, "Optimal fueling strategies for locomotive fleets in railroad networks," *Transp. Res. B, Methodol.*, vol. 44, nos. 8–9, pp. 1104–1114, Sep. 2010.
[34] T. Raviv and M. Kaspi, "The locomotive fleet fueling problem," *Oper. Res. Lett.*, vol. 40, no. 1, pp. 39–45, Jan. 2012.
[35] T. Papalexopoulos, C. Tjandraatmadja, R. Anderson, J. P. Vielma, and D. Belanger, "Constrained discrete black-box optimization using mixed-integer programming," in *Proc. ICML*, Jan. 2021, pp. 17295–17322. [Online]. Available: https://proceedings.mlr.press/v162/papalexopoulos22a.html
[36] M. Gasse, D. Chételat, N. Ferroni, L. Charlin, and A. Lodi, "Exact combinatorial optimization with graph convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 1–13. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2019/file/d14c2267d848abeb81fd590f371d39bd-Paper.pdf
[37] T. Q. T. Vo, M. Baiou, V. H. Nguyen, and P. Weng, "Learning to cut generation in branch-and-cut algorithms for combinatorial optimization," *ACM Trans. Evol. Learn. Optim.*, pp. 1–27, Apr. 2025, doi: 10.1145/3728371.
[38] R. Nellikkath and S. Chatzivasileiadis, "Physics-informed neural networks for AC optimal power flow," *Electric Power Syst. Res.*, vol. 212, Nov. 2022, Art. no. 108412. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0378779622005636
[39] L. Kong, J. Cui, Y. Zhuang, R. Feng, B. A. Prakash, and C. Zhang, "End-to-end stochastic optimization with energy-based model," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 11341–11354. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/file/49cf35ff2298c10452db99d08036805b-Paper-Conference.pdf
[40] C. Eisenmann and P. Plötz, "Two methods of estimating long-distance driving to understand range restrictions on EV use," *Transp. Res. D, Transp. Environ.*, vol. 74, pp. 294–305, Sep. 2019.
[41] C. De Cauwer, W. Verbeke, T. Coosemans, S. Faid, and J. Van Mierlo, "A data-driven method for energy consumption prediction and energy-efficient routing of electric vehicles in real-world conditions," *Energies*, vol. 10, no. 5, p. 608, May 2017.
[42] H. Mediouni, A. Ezzouhri, Z. Charouh, K. El Harouri, S. El Hani, and M. Ghogho, "Energy consumption prediction and analysis for electric vehicles: A hybrid approach," *Energies*, vol. 15, no. 17, p. 6490, Sep. 2022.
[43] B. Ruszczak, A. M. Wijata, and J. Nalepa, "Unbiasing the estimation of chlorophyll from hyperspectral images: A benchmark dataset, validation procedure and baseline results," *Remote Sens.*, vol. 14, no. 21, p. 5526, Nov. 2022.
[44] B. Ruszczak, K. Smykała, and K. Dziubański, "The detection of Alternaria solani infection on tomatoes using ensemble learning," *J. Ambient Intell. Smart Environ.*, vol. 12, no. 5, pp. 407–418, Sep. 2020.
[45] B. Ruszczak and D. Boguszewska-Mańkowska, "Soil moisture a posteriori measurements enhancement using ensemble learning," *Sensors*, vol. 22, no. 12, p. 4591, Jun. 2022.
[46] D. Boguszewska-Mańkowska, B. Ruszczak, and K. Zarzyńska, "Classification of potato varieties drought stress tolerance using supervised learning," *Appl. Sci.*, vol. 12, no. 4, p. 1939, Feb. 2022.
[47] B. Ruszczak, K. Kotowski, J. Andrzejewski, A. Musiał, D. Evans, V. Zelenevskiy, S. Bammens, R. Laurinovics, and J. Nalepa, "Machine learning detects anomalies in OPS-SAT telemetry," in *Computational Science ICCS*. Cham, Switzerland: Springer, 2023, pp. 295–306.
[48] B. Ruszczak, "Evolutionary configuration of the temporal convolutional network for the electricity price prediction," in *Proc. Companion Conf. Genetic Evol. Comput.*, Jul. 2023, pp. 75–76.
[49] The Electric Circuit by Hydro-Quebec. *Vehicle Characteristics and Impact on Charging*. Accessed: Mar. 15, 2025. [Online]. Available: https://lecircuitelectrique.com/en/charging/
[50] H. P. Williams, *Model Building in Mathematical Programming*, 5th ed., Chichester, U.K.: Wiley, 2013.
[51] J. T. Betts, *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming* (Advances in Design and Control), vol. 19, 2nd ed., Philadelphia, PA, USA: Society for Industrial and Applied Mathematics, 2010.
[52] W. Ogryczak and T. Śliwiński, "Decision support under risk by optimization of scenario importance weighted OWA aggregations," *J. Telecommun. Inf. Technol.*, vol. 3, no. 3, pp. 5–13, Sep. 2009.
[53] P. Jaillet, J. Qi, and M. Sim, "Routing optimization under uncertainty," *Oper. Res.*, vol. 64, no. 1, pp. 186–200, Feb. 2016.
[54] T. Aven, "An integrated framework for decision support on risk and uncertainty," *Risk Manage.*, vol. 12, no. 4, pp. 285–300, Oct. 2010.
[55] W. Ogryczak, "Reference point method with importance weighted partial achievements," *J. Telecommun. Inf. Technol.*, vol. 2, no. 4, pp. 17–25, Dec. 2008.

PIOTR PAŁKA received the M.Sc. (Eng.) and Ph.D. degrees in computer science from the Faculty of Electronics and Information Technology (FEIT), Warsaw University of Technology (WUT), Poland, in 2005 and 2009, respectively. He is an Assistant Professor with the Institute of Control and Computation Engineering (ICCE), WUT. He is the author/co-author of over 100 conferences and journal articles and books. His research interests include multi-agent systems, communication problems, agent-based modeling systems, and the development of complex systems.

TOMASZ ŚLIWIŃSKI received the M.Sc. and Ph.D. degrees in computer science from Warsaw University of Technology, Poland, in 1999 and 2006, respectively. Since 2006, he has been held the position of an Assistant Professor with the Institute of Control and Computation Engineering, Warsaw University of Technology. His research interests include optimization, continuous and discrete, linear and nonlinear, with single and multiple criteria, applied to business and industrial processes. His current focus is on the application of neural networks in discrete optimization and exploring new methods to enhance computational efficiency.

PRZEMYSŁAW KASZYŃSKI received the M.Sc. (Eng.) degree, specializing in sustainable energy development and the Ph.D. degree in mining engineering and geology from the AGH University of Science and Technology, Kraków, Poland, in 2009 and 2018, respectively. He is an Assistant Professor with the Division of Energy Economics, Mineral and Energy Economy Research Institute, Polish Academy of Sciences. His research interests include the development of models for economic, environmental, and technological relations in fuel and energy industries.

MARTA KUTA received the degree from the Faculty of Energy and Fuels, AGH University of Science and Technology (AGH UST), Kraków, Poland. She has been professionally affiliated with AGH UST, as a Research and Teaching Assistant, since 2014. She also has experience in international environments that foster innovation. Further, she honed her expertise through internships with the University of Cambridge, the University of Oxford, and Dublin Institute of Technology. Her research interests are primarily focused on the production, storage, and management of electrical and thermal energy.

BOGDAN RUSZCZAK (Member, IEEE) received the M.Sc. degree in management and the M.Sc. degree in computer science from the University of Technology, Opole, Poland, in 2005 and 2006, respectively, and the Ph.D. degree from Wrocław University of Technology, Poland, in 2012. He is a Machine Learning Specialist with a strong university background. He is an Assistant Professor with the Department of Informatics, Opole University of Technology, where he joined in 2005. His current research interests include image processing, hyperspectral image understanding, sensor measurement modeling, machine learning, and deep learning.

MARCIN MALEC received the M.Sc. degree in chemical sciences from AGH University, Kraków, Poland, in 2013, and the Ph.D. degree in environmental engineering, mining, and energy sciences from the Mineral and Energy Economy Research Institute (MEERI), Polish Academy of Sciences (PAS), Kraków, in 2022. He is an Assistant Professor with MEERI, PAS. He is the author or co-author of more than 20 papers in the field of mathematical modeling of energy and environmental systems. His research interests focus on energy and fuel market analysis and optimization tools. In his work, he uses mathematical modeling as a research method.

PIOTR SAŁUGA received the D.Sc. degree. Since 2020, he has been an Associate Professor with WSB University. From 2013 to 2020, he was with the AGH University of Science and Technology, Kraków. He is responsible for the implementation of research work in the field of management, decision-making process, and construction of models of fuel and energy systems. He is a Mineral Asset Appraiser and an Economist, specializing in risk analysis, stochastic simulation applications, and real options valuation, with over 20 years of experience. He is the author of over 50 research publications and monographs.

JACEK KAMIŃSKI is a Full Professor with the Mineral and Energy Economy Research Institute, Polish Academy of Sciences, Kraków, Poland. He was the Head of the Department of Energy Policy and Markets and the Division of Energy Economics. He has more than 25 years of experience in energy economics, energy markets, energy and climate policy, and mathematical modeling of energy systems. From 2007 to 2009, he was a Detached National Expert with the DG Joint Research Centre of the European Commission, Institute for Prospective Technological Studies in Seville, Spain. He is the author or co-author of more than 120 papers. He was a Former of the Polish affiliation of the International Association for Energy Economics.